

图的搜索问题

《算法设计与分析》

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02



深度优先搜索



二、深度优先搜索

- 深度优先搜索(Depth First Search, 简记为DFS)是一种递归算法, 它从一个顶点 v 出发, 然后依次对 v 的邻接顶点产生递归调用, 直到每个顶点都搜索完为止。深度优先搜索在执行的过程中, 对每个顶点 v 做标记, 每个顶点有两个标记, 一个是 $d[v]$, 记录第一次被访问的时间, 而第二个标记 $f[v]$ 记录当完成对邻接表搜索的时间。

DFS(G)

```
1  for each vertex  $u \in V$  do
2       $color[u] \leftarrow \text{White}$ 
3       $\pi[u] \leftarrow \text{NIL}$ 
4   $time \leftarrow 0$ 
5  for each vertex  $u \in V$  do
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7          DFSVisit( $u$ )
```

Running time:
 $O(|V| + |E|)$

DFSVisit(u)

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An example

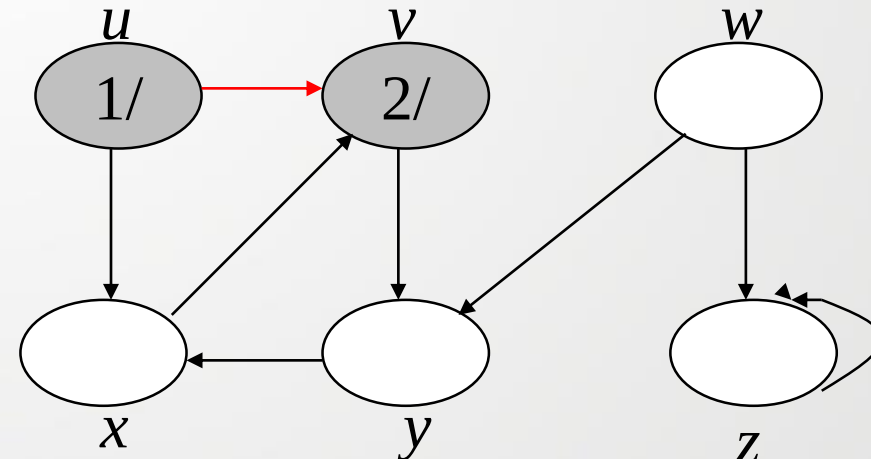
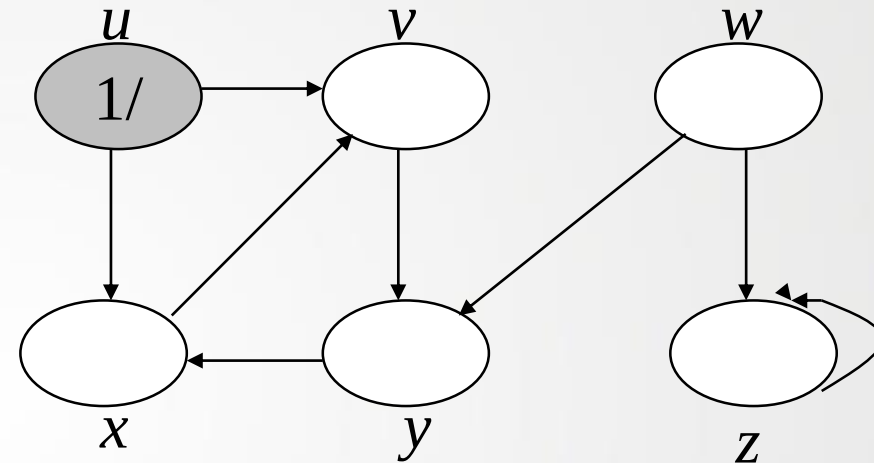
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$$d[v]/f[v]$$



An example

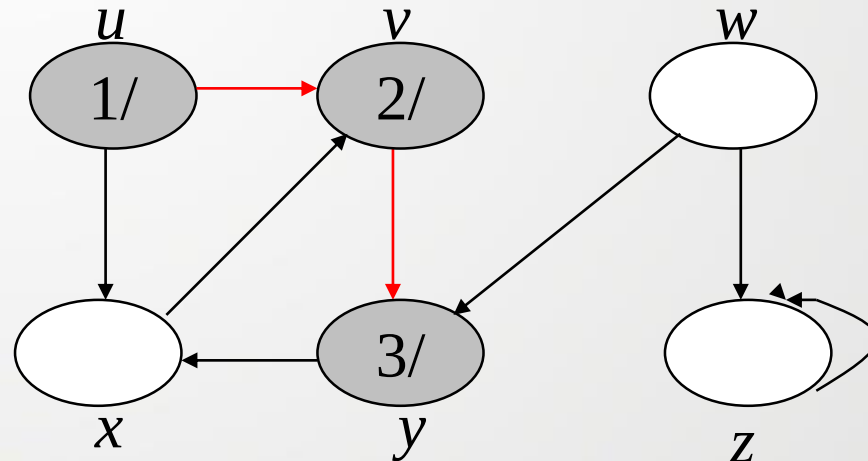
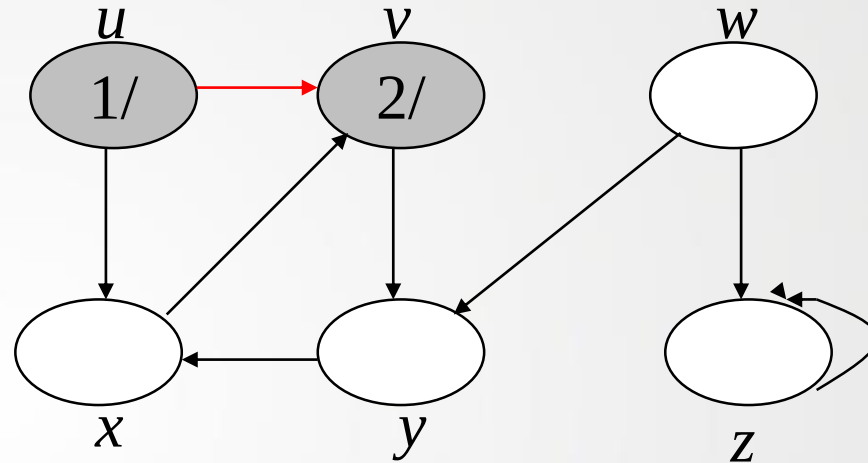
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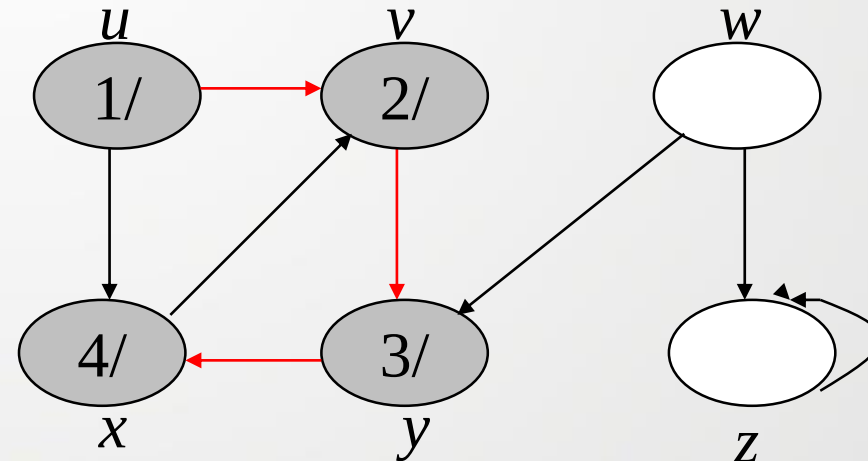
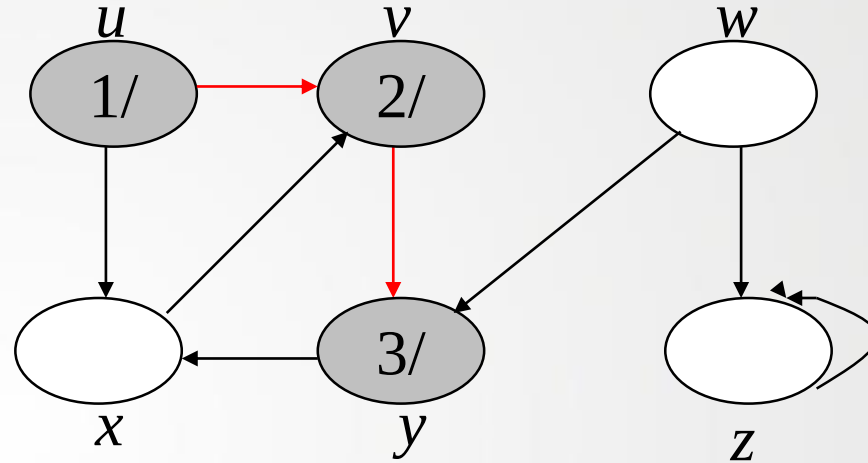
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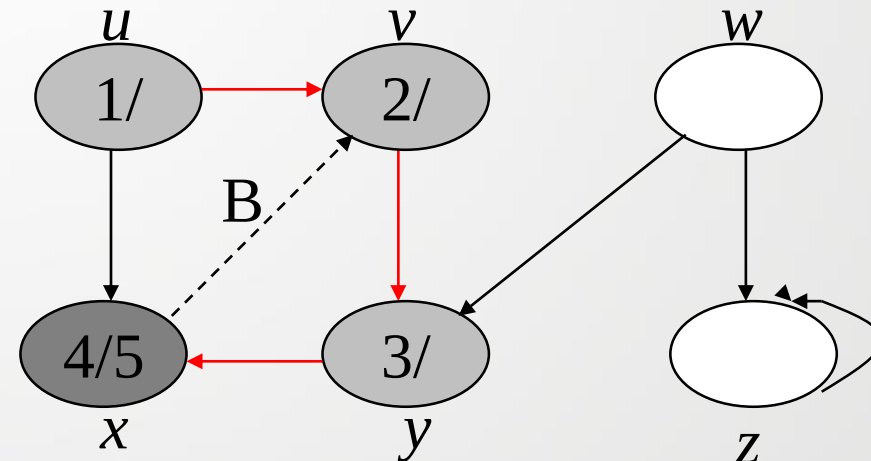
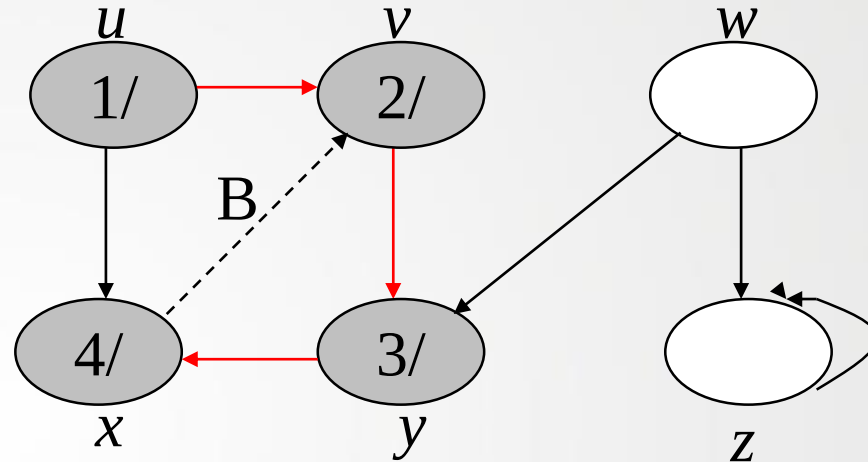
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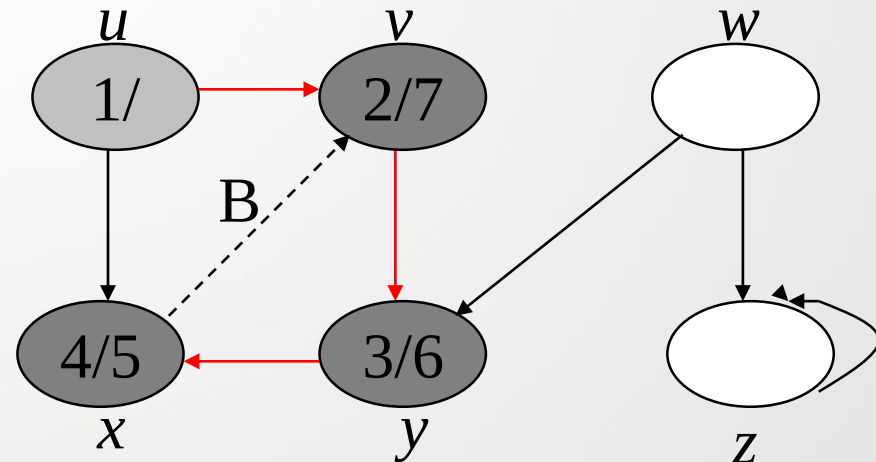
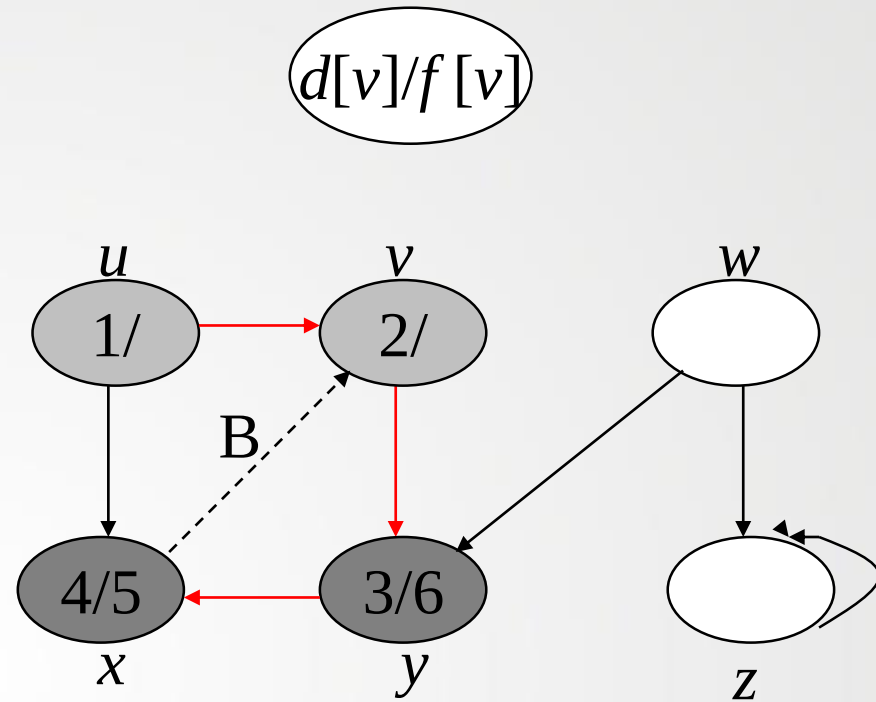
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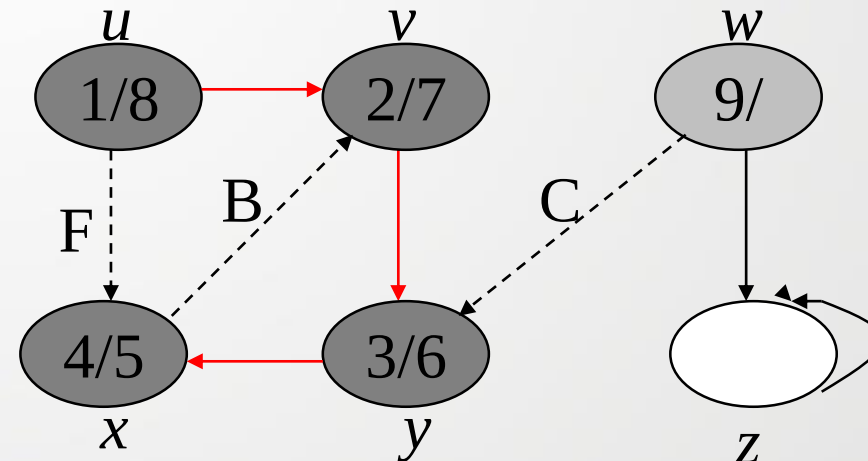
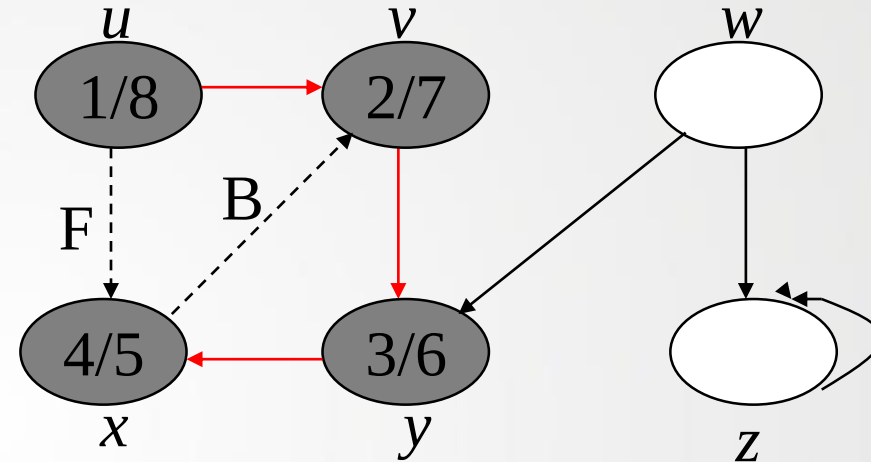
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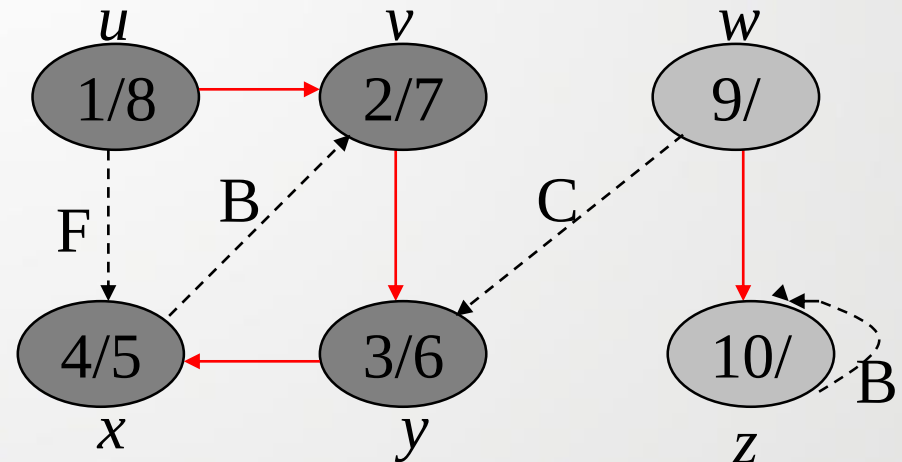
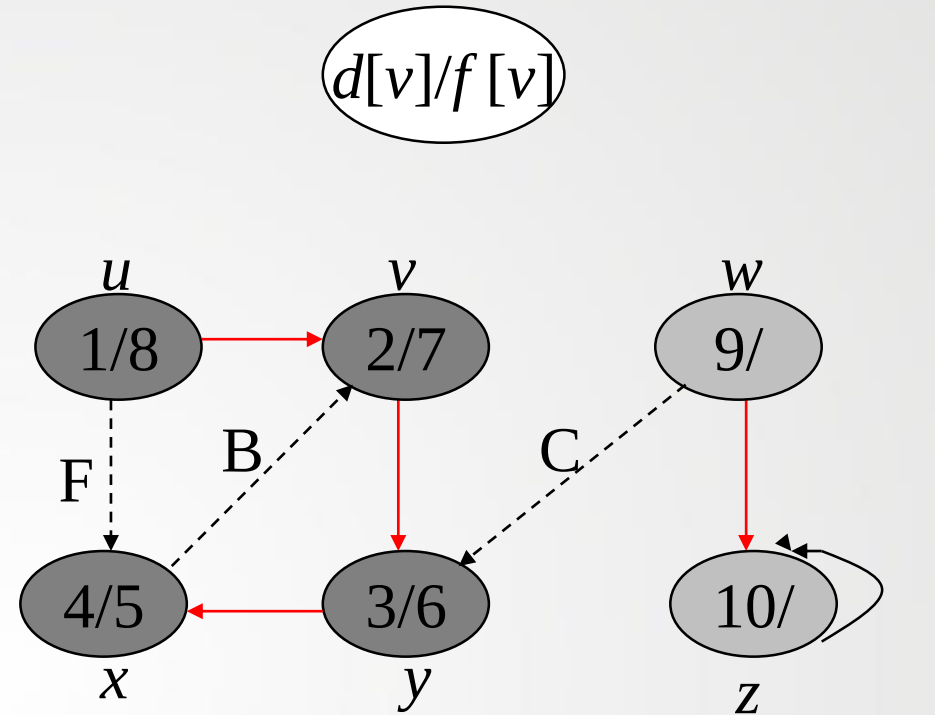
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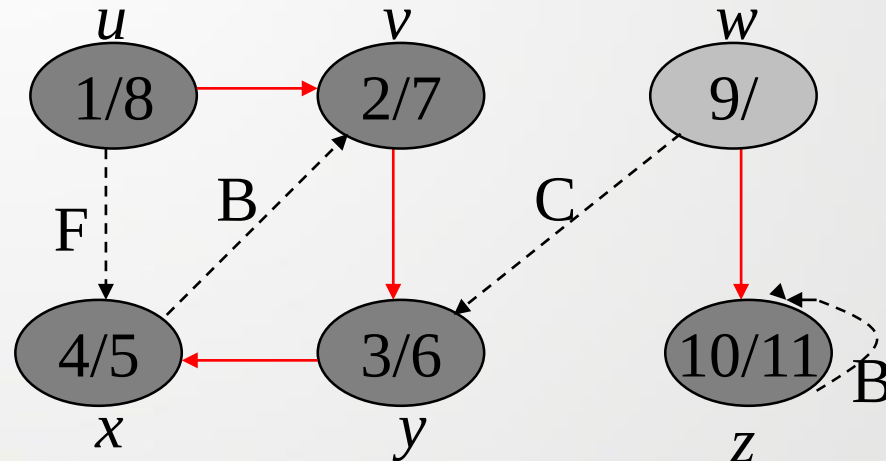
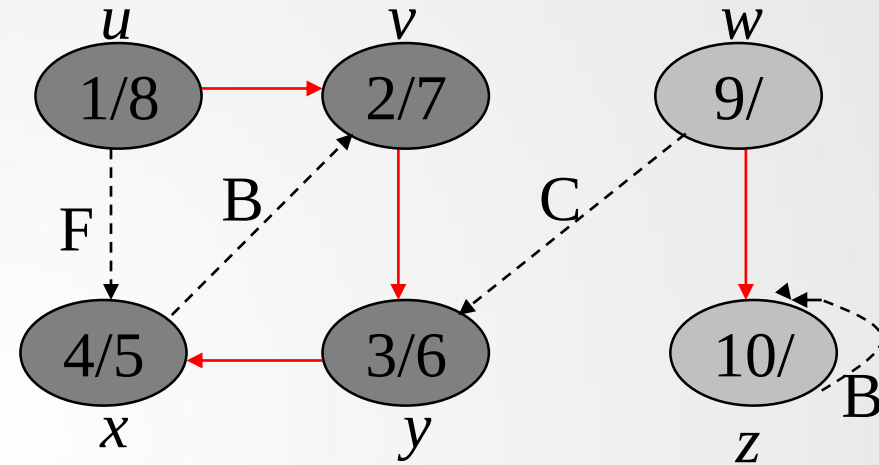
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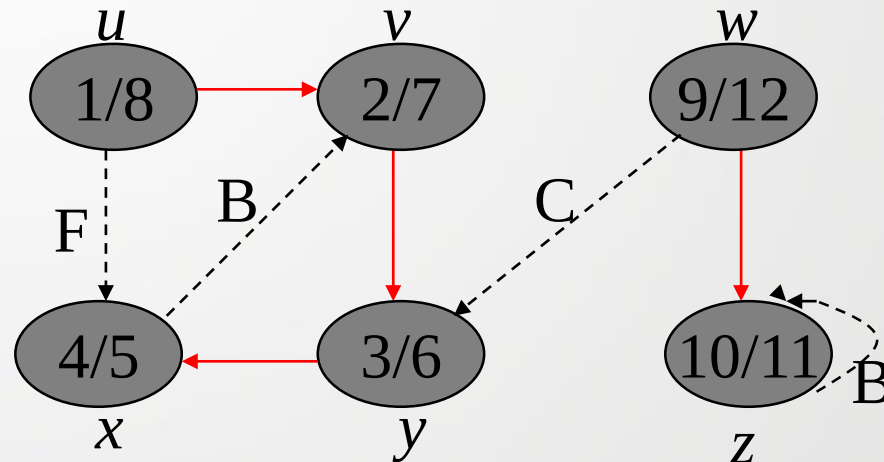
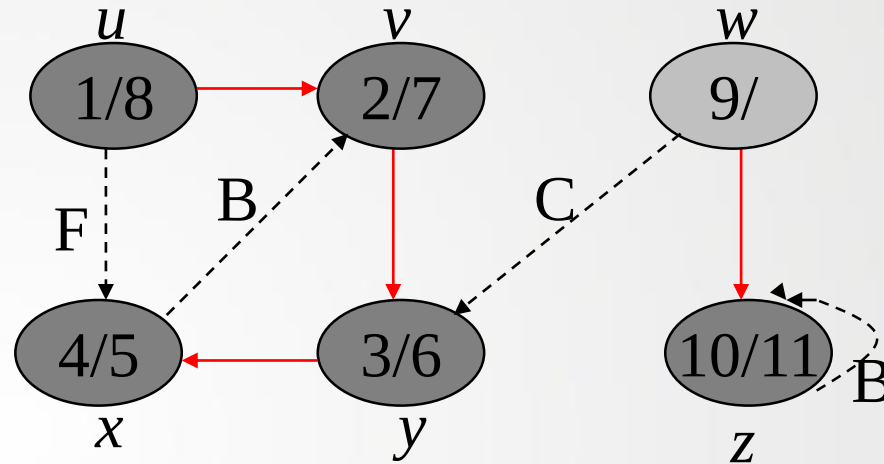
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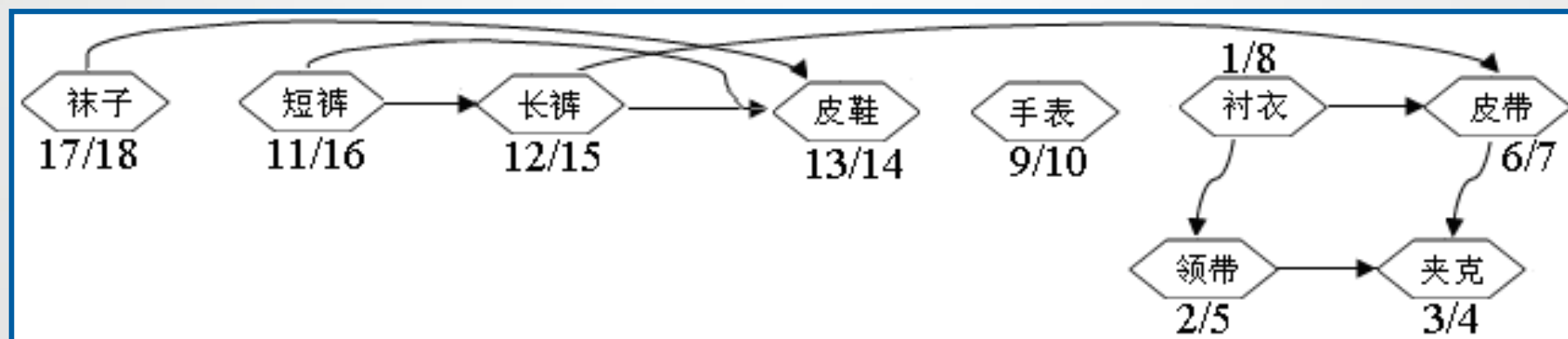
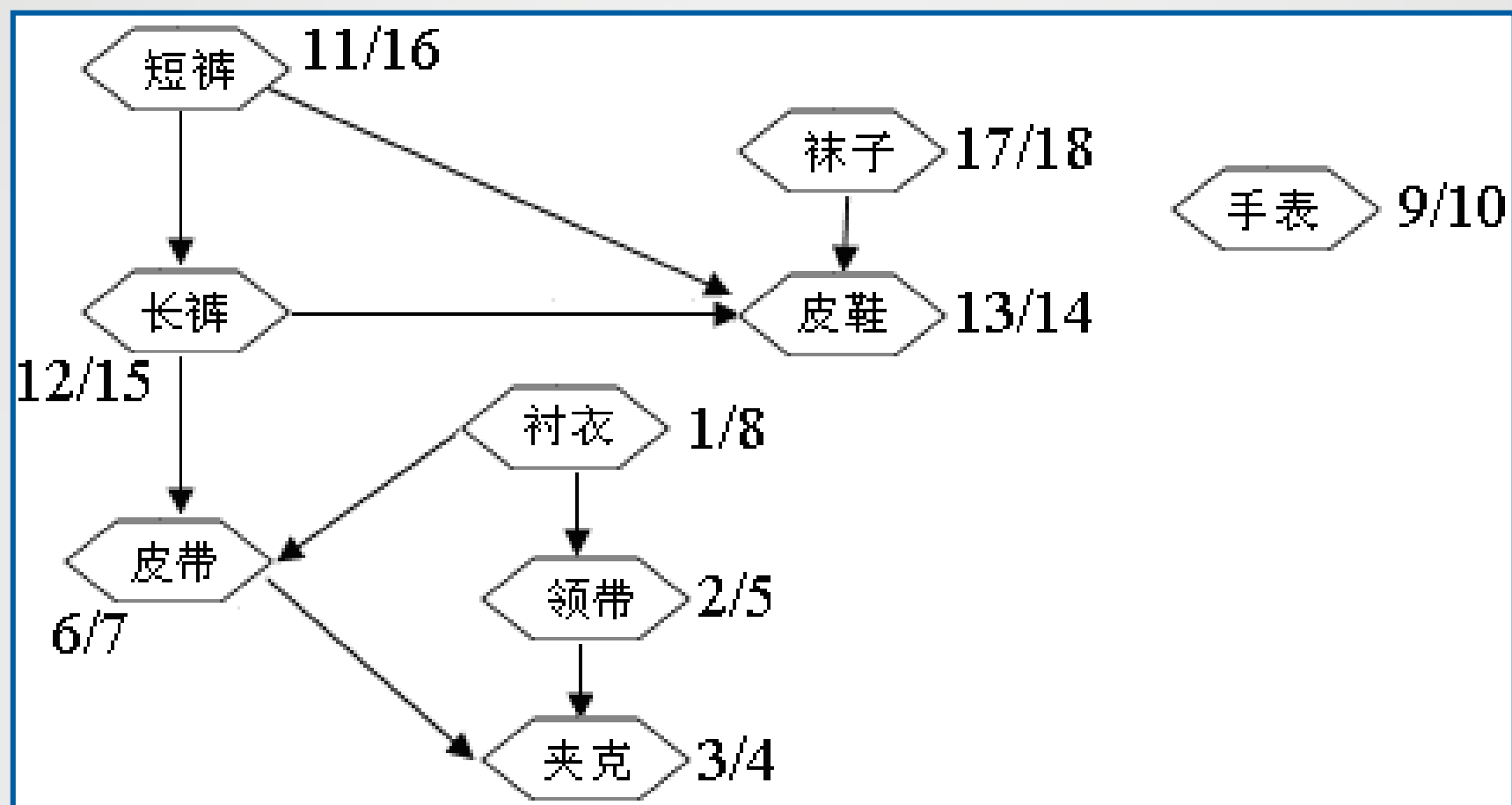
有趣的性质

- **深度优先搜索**的子图 G_π 构成的是一座森林，它由许多深度优先树构成。
- 深度优先搜索完成后，将有向图的边分成四类：
 - ① 属于深度优先树中的边，常称为树边；
 - ② 不在深度优先树中，但是由树中的顶点指向其父顶点或者是指向顶点本身的边，常称为后向边(B)；
 - ③ 不在深度优先树中，但是由树中的顶点指向其子辈顶点的边，常称为前向边(F)；
 - ④ 其他的边统称为交叉边(C)



应用1： 拓扑排序

- 给定一个有向非循环图 $G = (V, E)$ ，图 G 的拓扑排序是所有顶点的一个线性排序，它使得如果图 G 包含边 (u, v) ，那么在该顶点序列中， u 一定在 v 的前面。图 G 的拓扑排序能够看作水平方向的一个顶点序列，使得所有的有向边从左指向右。它可以用来表示事件之间的优先关系。





应用1: 拓扑排序

TopologicalSort(G)

- 1 call **DFS(G)** to compute finishing times $f[v]$ for each vertex v
- 2 as each vertex is finished, insert it onto the front of a linked list
- 3 **return** the linked list of vertices

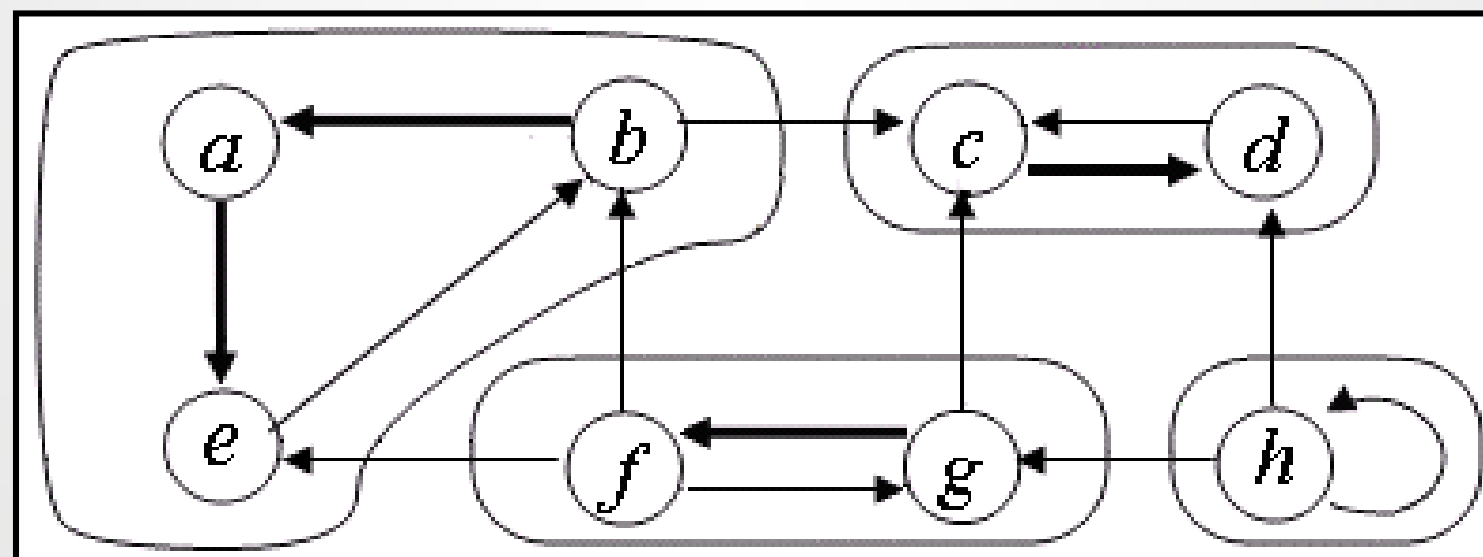
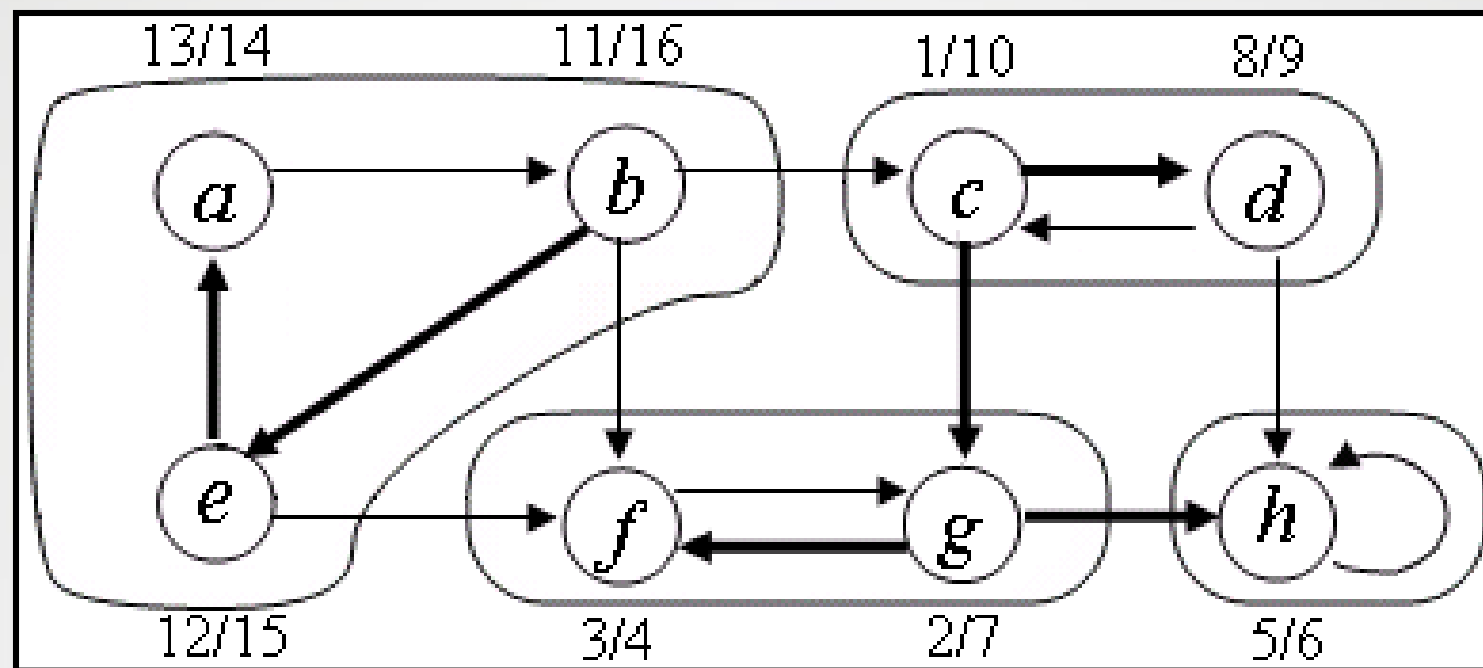


应用2：强连通分支

- 有向图 $G = (V, E)$ 的一个强连通分支是一个最大顶点集 $C \subseteq V$, 使得对于 C 中的每一对顶点 u 和 v , 它们都互相可达, 即 u 可以到 v , v 也可以到 u 。图8.6给出了一个例子, 图中用圈圈起来的顶点集均构成连通分支。

StrongConnectedComponents(G)

- 1 call DFS(G) to compute finishing times $f[u]$ for each vertex u**
- 2 compute G^T**
- 3 call DFS (G^T), but in the main loop of DFS, consider the vertices in order of decreasing $f[u]$ (as computed in line 1)**
- 4 output the vertices of each tree in the depth-first forest formed in line 3 as a separate strongly connected component**



谢谢大家

